Supporting Information

Unveiling Molecular Interactions in Chocolate by Molecular Dynamics: A Study on Caffeine, Fatty Acids and Sugars

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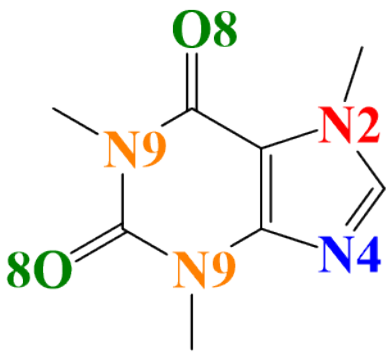


Figure S1. Schematic representation of caffeine

Table S1. System details

Solvent	Acesulfame-K	Maltose	Mannitol
Initial box dimension(nm^3)	12 × 12 × 12	12 × 12 × 12	12 × 12 × 12
Final box dimension(nm^3)	7.06 × 7.06 × 7.06	6.52 × 6.52 × 6.52	6.52 × 6.52 × 6.52
Total number of sugar molecules	930	310	310
Total number of caffeine molecules	15	15	15
Total number of linoleic acid molecules	17	17	17
Total number of oleic acid molecules	178	178	178
Total number of palmitic acid molecules	39	39	39
Total number of stearic acid molecules	58	58	58

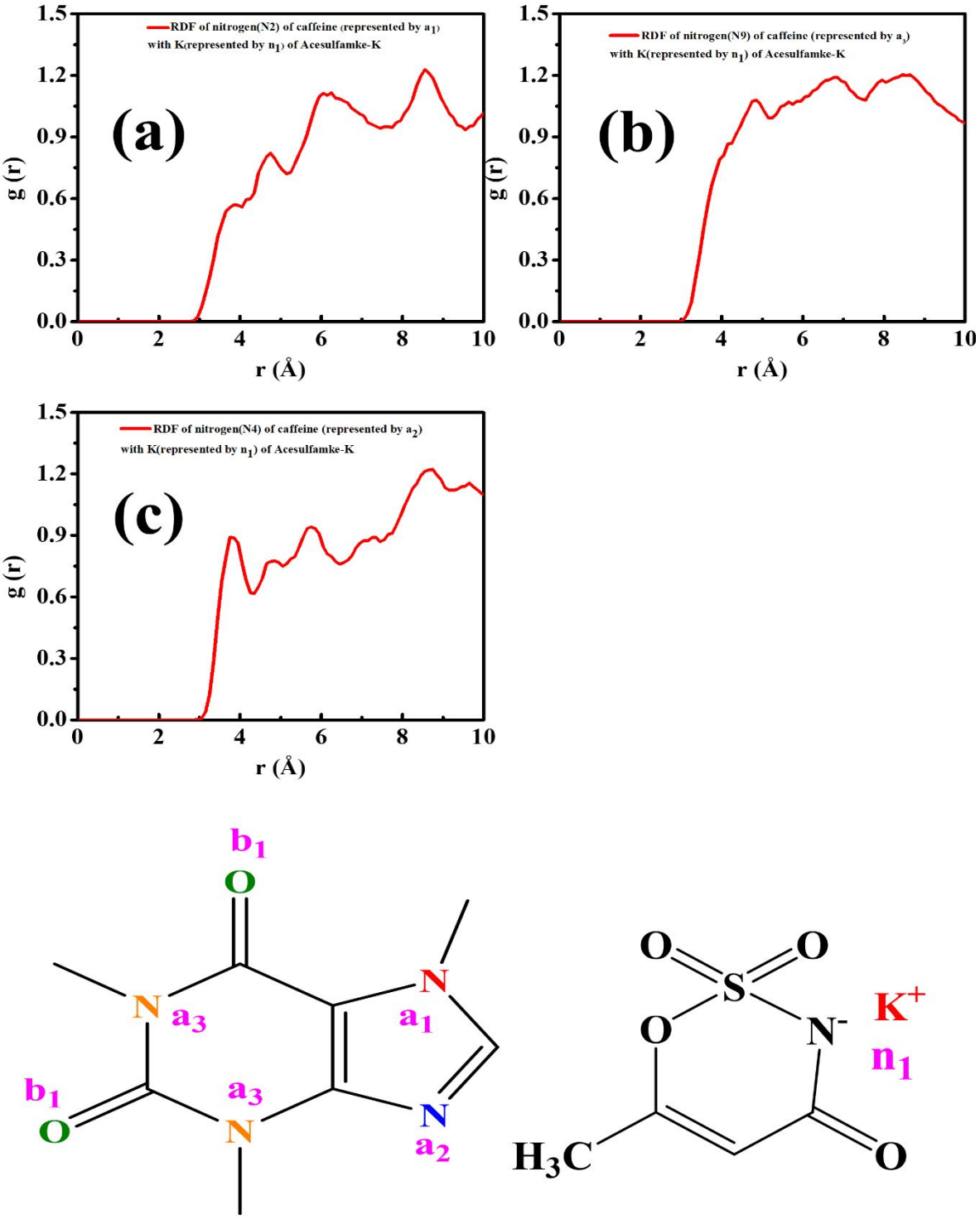


Figure S2. RDF of caffeine in acesulfame-K (a) (denoted by $a_1 \cdots n_1$) (b) N9-K (denoted by $a_3 \cdots n_1$) and (c) N4-K(denoted by $a_2 \cdots n_1$)

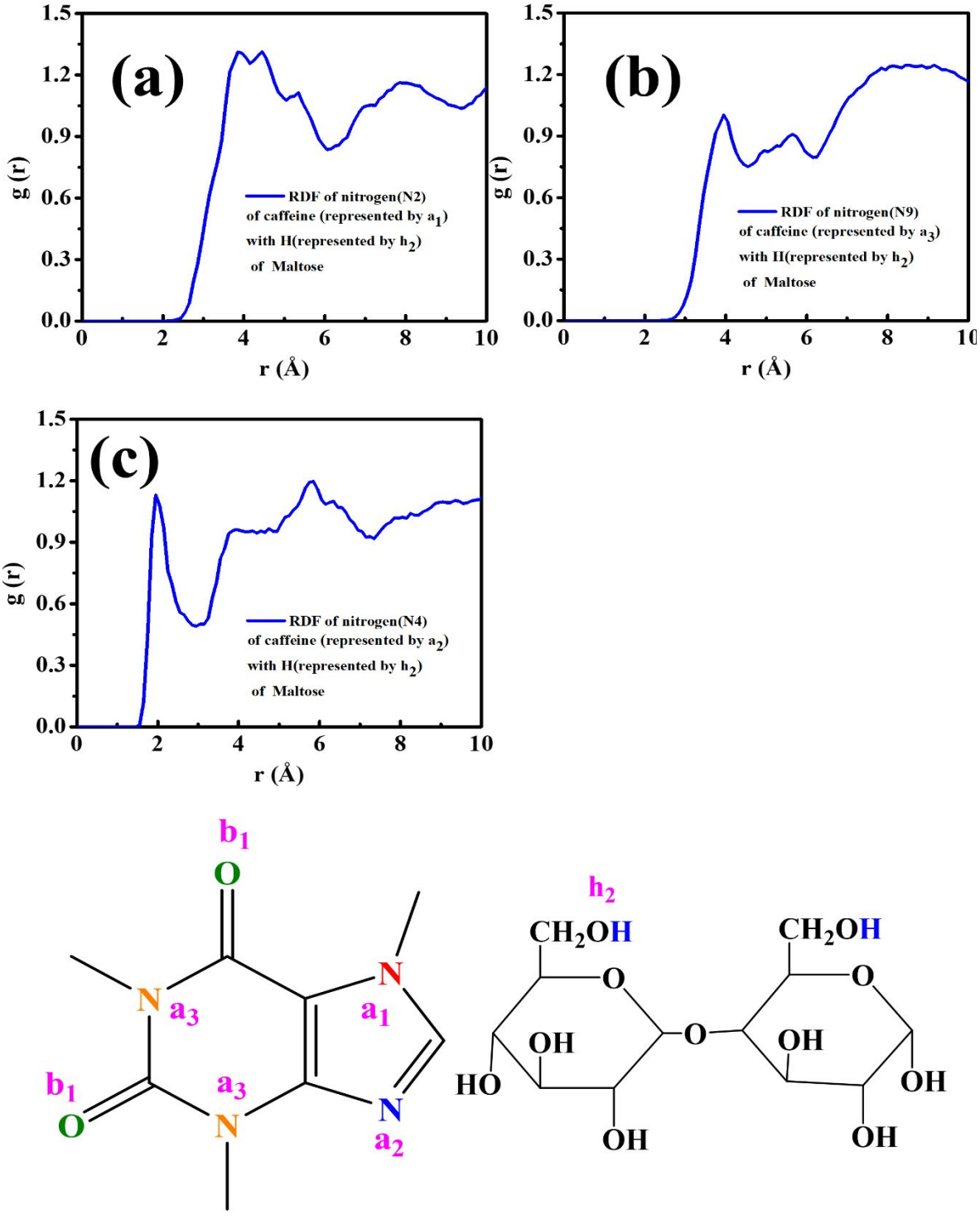


Figure S3. RDF of caffeine in Maltose (a) N2-H (denoted by $a_1 \cdots h_2$) (b) N9-H (denoted by $a_2 \cdots h_2$) and (c) N4-H (denoted by $a_3 \cdots h_2$)

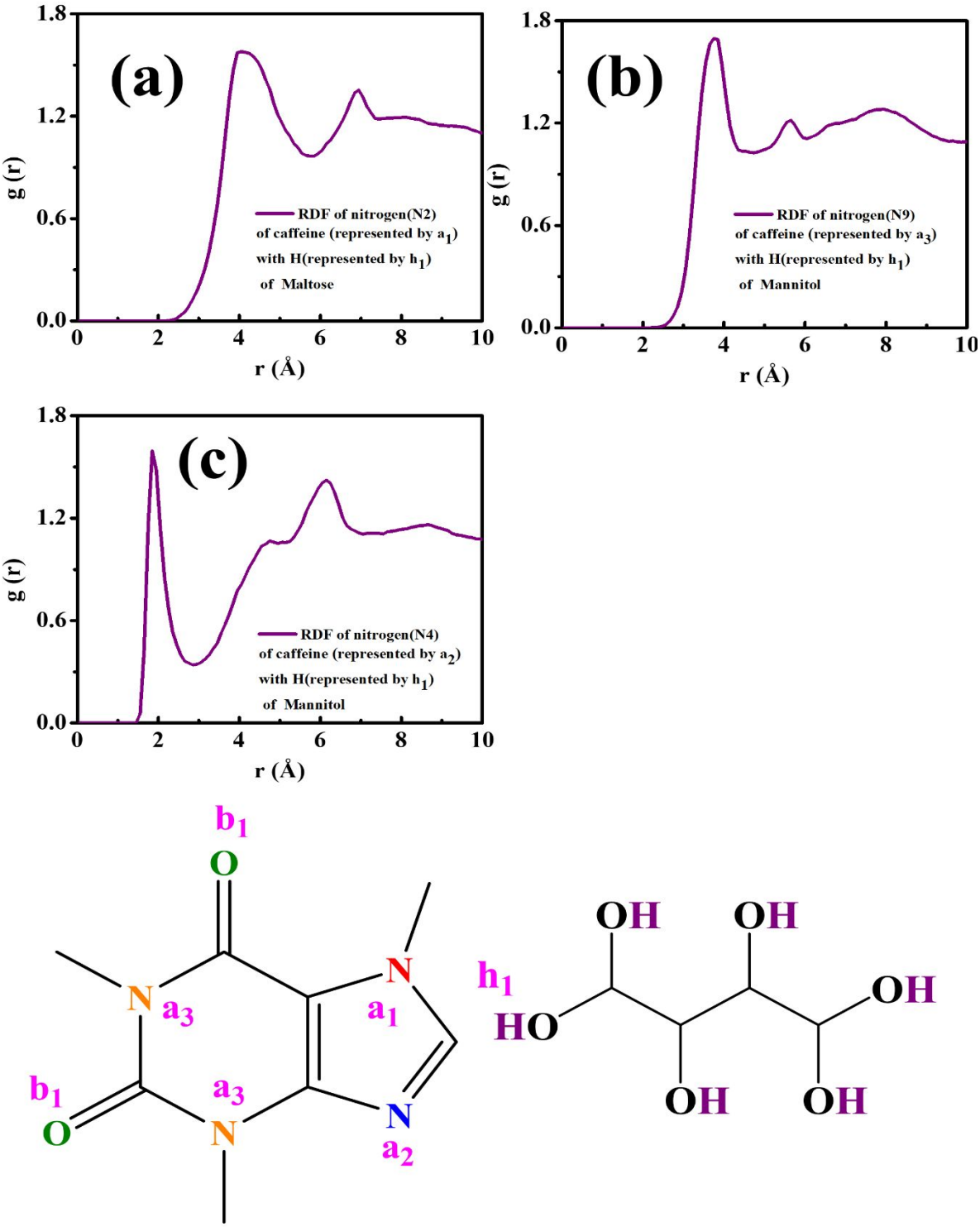


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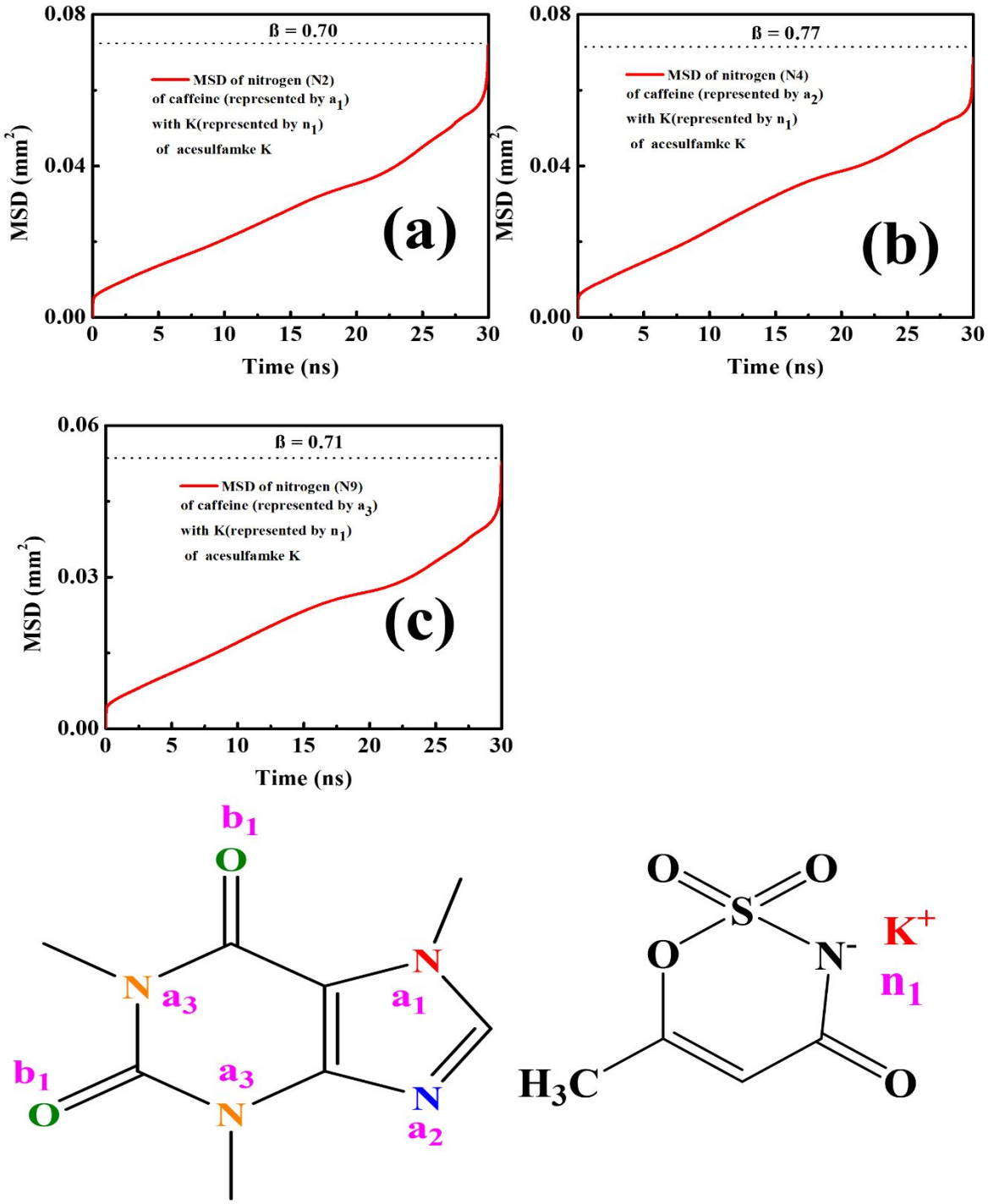


Figure S5. MSD of caffeine in acesulfame-K (a) N2-K (denoted by $a_1 \ldots n_1$) (b) N4-K (denoted by $a_2 \ldots n_1$) (c) N9-K(denoted by $a_3 \ldots n_1$)

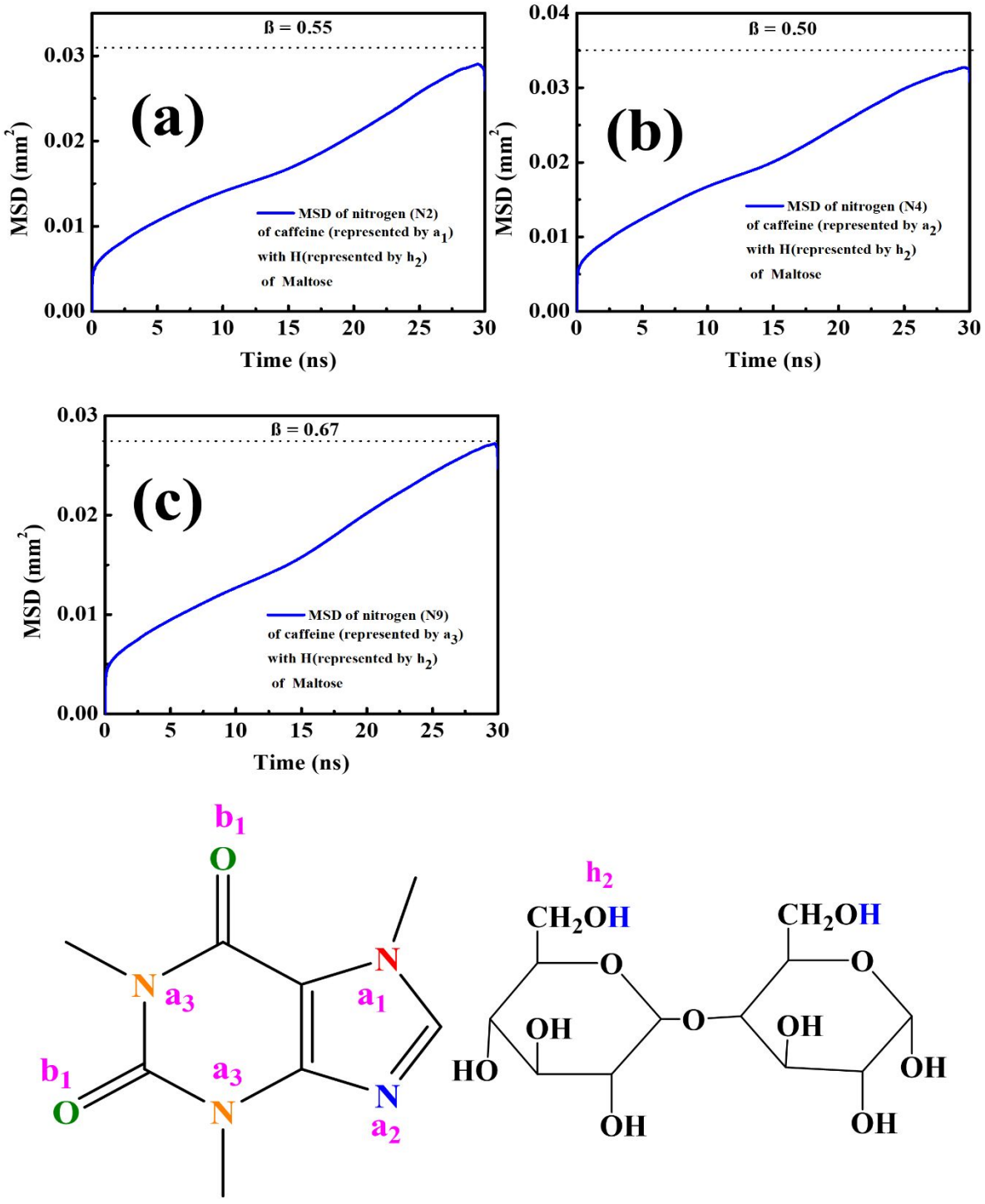


Figure S6. MSD of caffeine in Maltose (a) N2-H (denoted by $a_1 \cdots h_2$) (b) N4-H (denoted by $a_2 \cdots h_2$) (c) N9-H (denoted by $a_3 \cdots h_2$)

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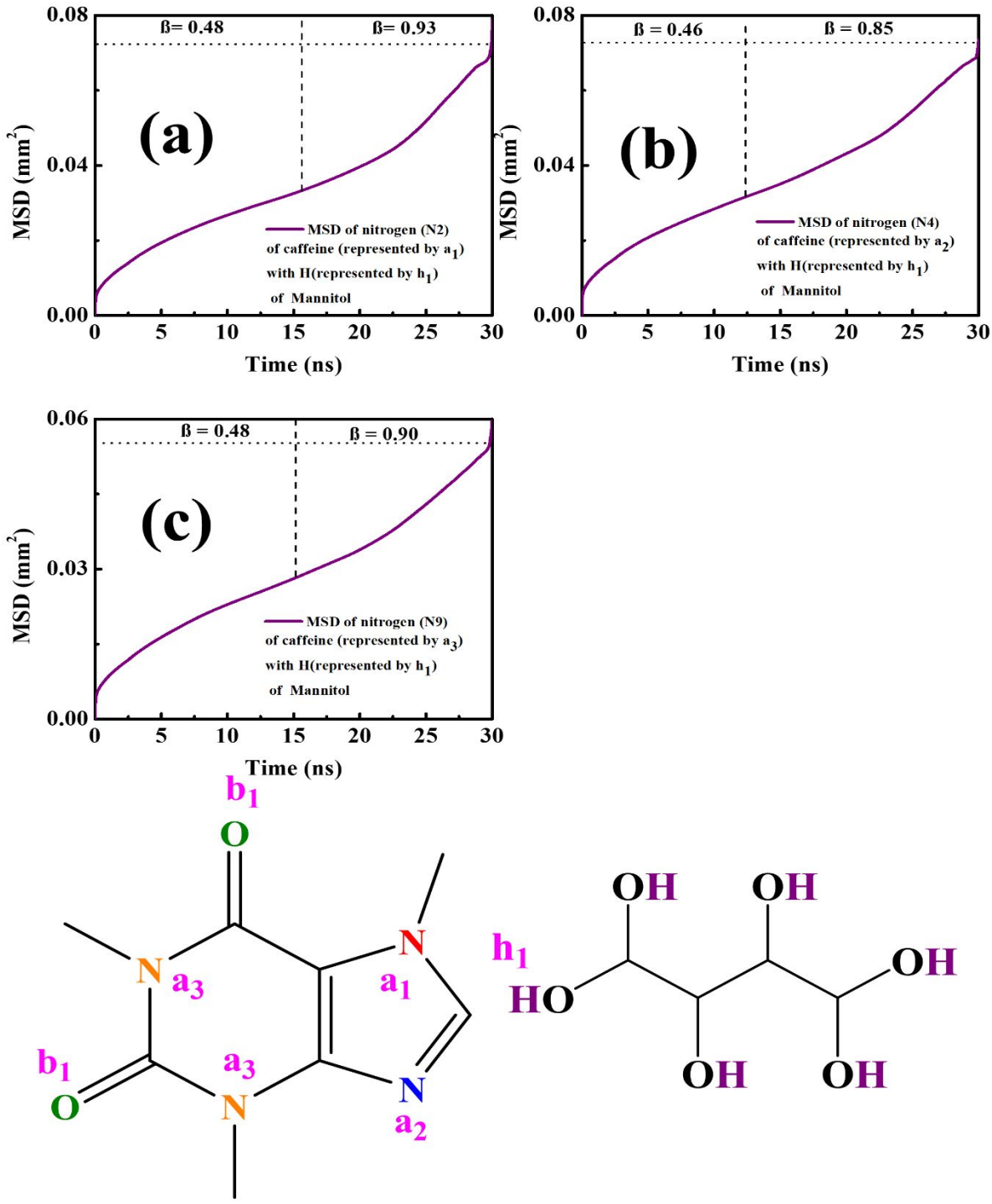


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